Systems of inequalities

1 **A, C, B**

2

a Dashed line **b** Solid line **c** Solid line **d** Dashed line

e Dashed line **f** Solid line **g** Solid line **h** Dashed line

3

a No **b** No **c** Yes **d** No

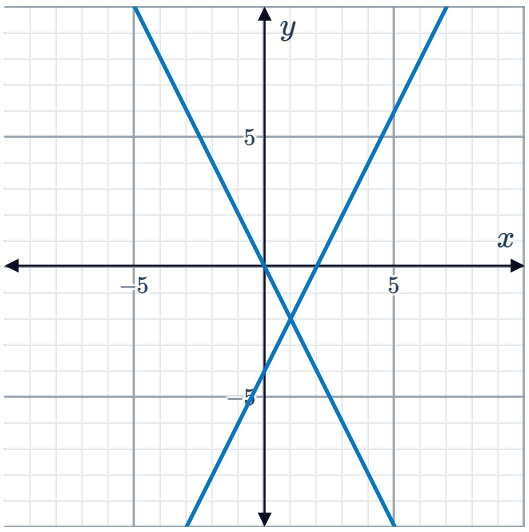
4 The set of a ordered pairs that satisfy each inequality in the system.

5

a

	$2x - y = 4$	$6x + 3y = 0$
x-intercept	$(2, 0)$	$(0, 0)$
y-intercept	$(0, -4)$	$(0, 0)$

b

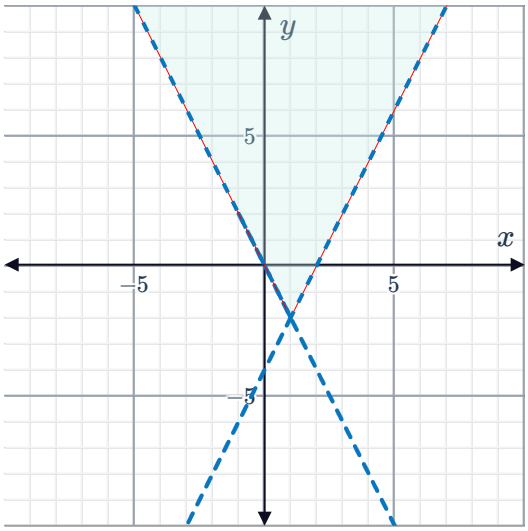


c

i No **ii** Yes **iii** No **iv** No

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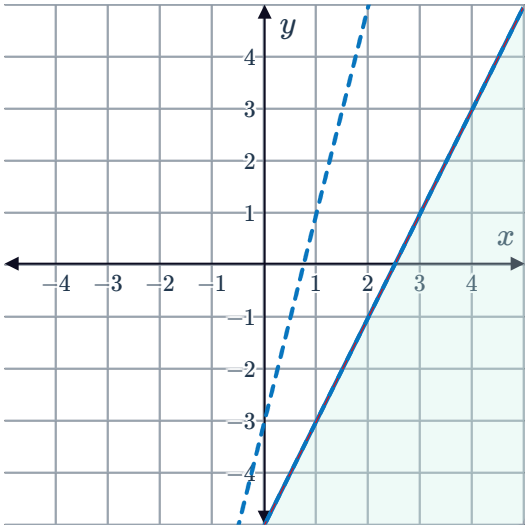
d



6

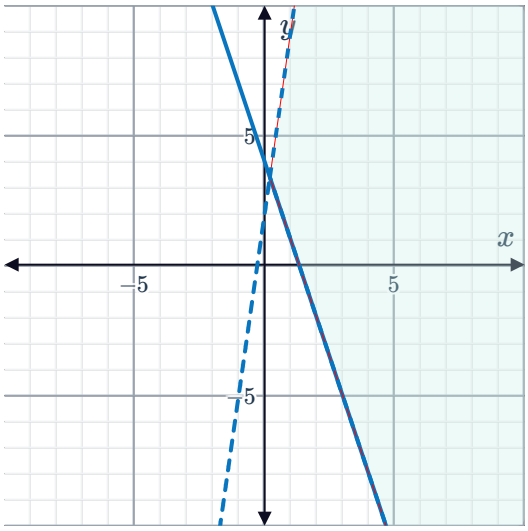
a $(-1, -7)$

b



7

a



b

i No

ii No

iii No

iv No

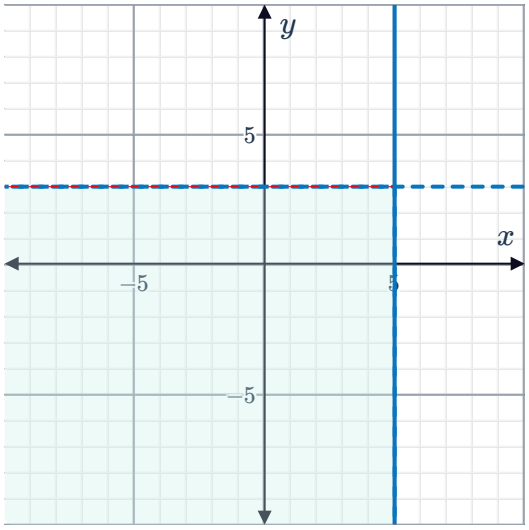
v Yes

vi Yes

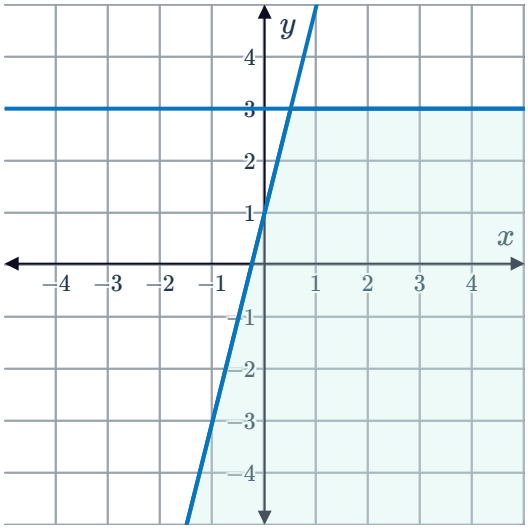
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8

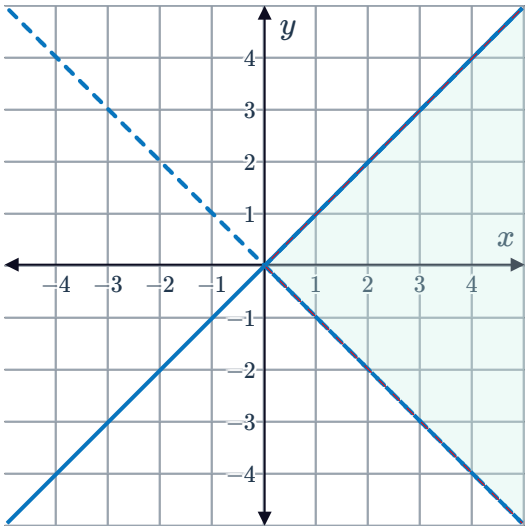
a



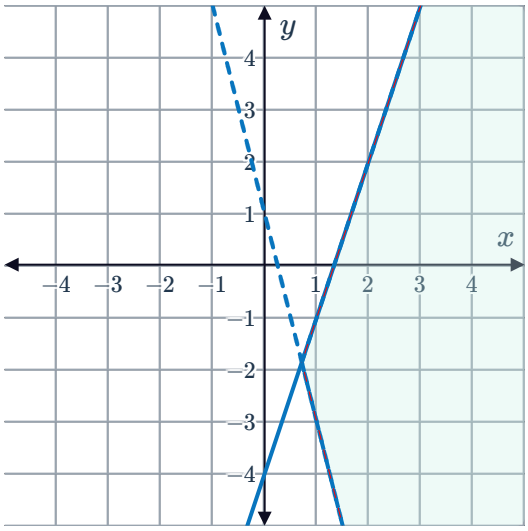
b



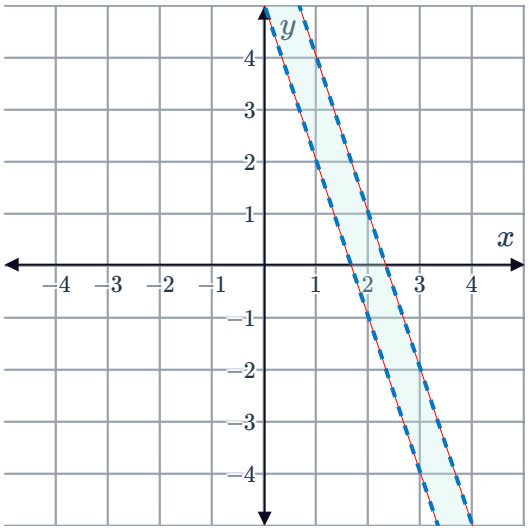
c



d



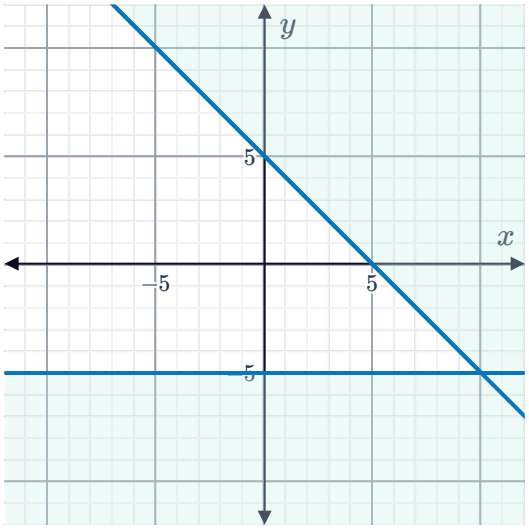
e



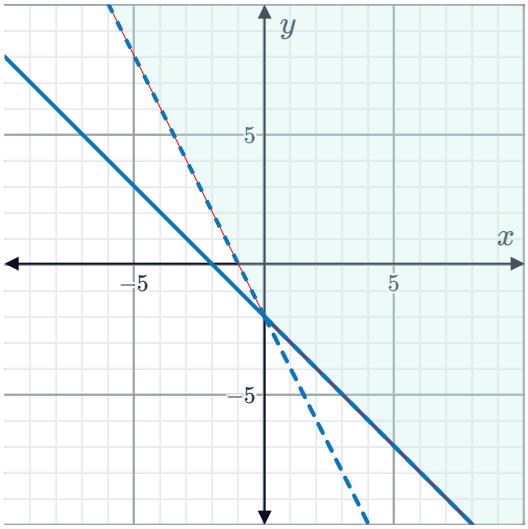
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9

a



b



10

a No solutions

b Infinitely many solutions

11

a $\begin{cases} x < 1 \\ y \leq 3 \end{cases}$

b $\begin{cases} y \geq 2x - 3 \\ y \leq 1 \end{cases}$

c $\begin{cases} y > x \\ y \leq -x \end{cases}$

d $\begin{cases} y + 3x > 0 \\ y + 3x \leq 3 \end{cases}$

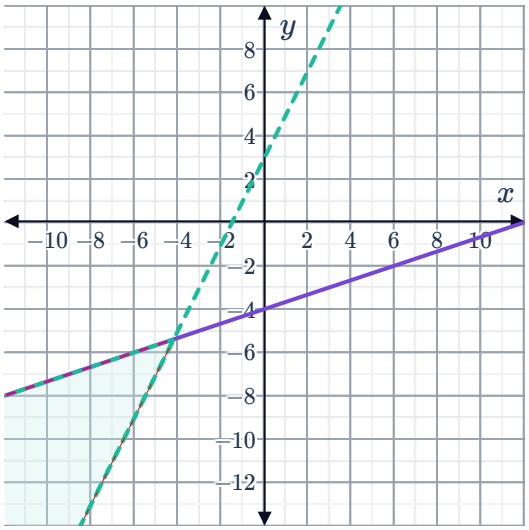
e $\begin{cases} x < 1 \\ y < 1 \\ y \geq -x - 5 \end{cases}$

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Additional questions

12

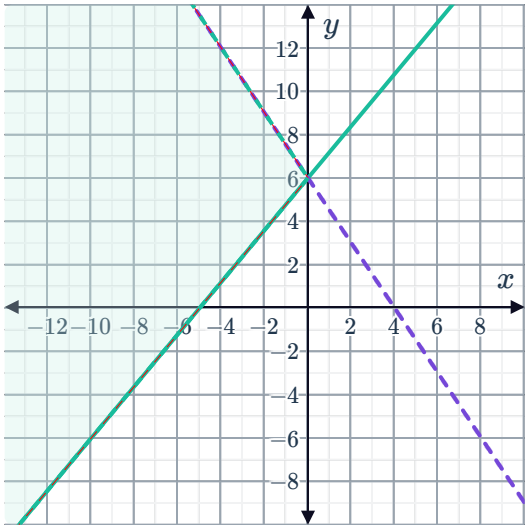
a



b Yes

13

a



b No

14

a $x \geq 3, y < \frac{1}{3}x - 1$

No, since both systems include at least one strict inequality, their point of intersection is not a valid solution.

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15 $x + y \leq 15, 0.8x \leq 32$

16 $b + m \leq 27, 10b + 13m \geq 260$

17

a $2v \leq m, 0.6v \leq 42$

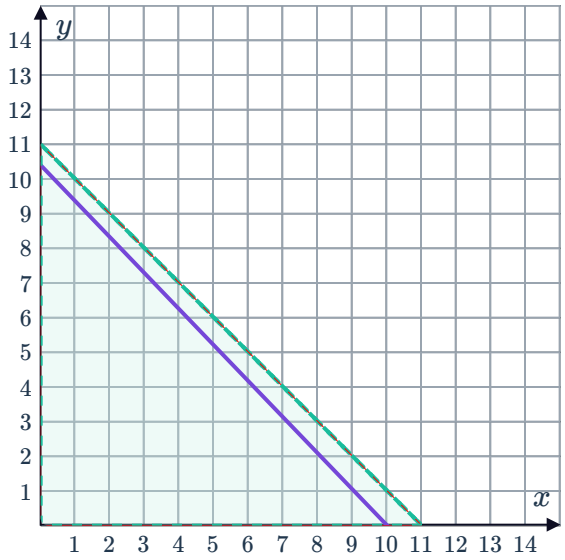
b Yes, since 28 vegan pizzas and 70 meat lovers pizzas would satisfy the constraints, and pizzas can only be made in whole numbers.

c No. While $(25.25, 71)$ satisfies the constraints, you cannot cook a fraction of a pizza.

18

a $w + m \leq 11, 2.6w + 2.5m < 26$

b



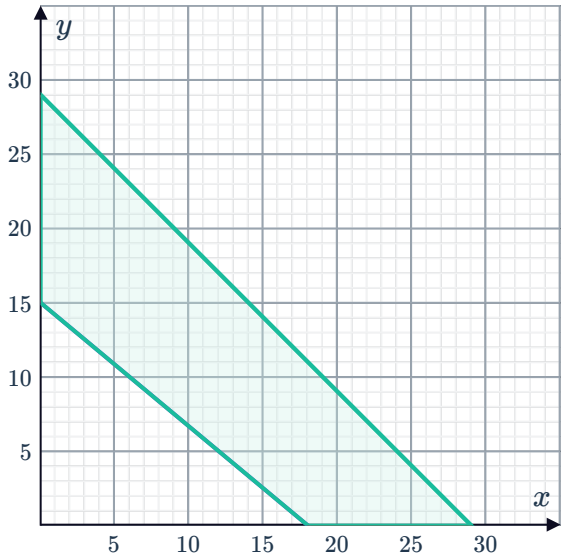
c Viable solution: Any point which satisfies the constraints and is made up of positive integers, since the variables represent the number of men and women. For example, $(3, 4)$.
Nonviable solution: Any point which includes at least one non integer value, since you cannot have part of a person as a team member. For example, $(3.1, 4)$.

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19

a $x + y \leq 29, 15x + 18y \geq 270$

b



c 10

d Yes

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